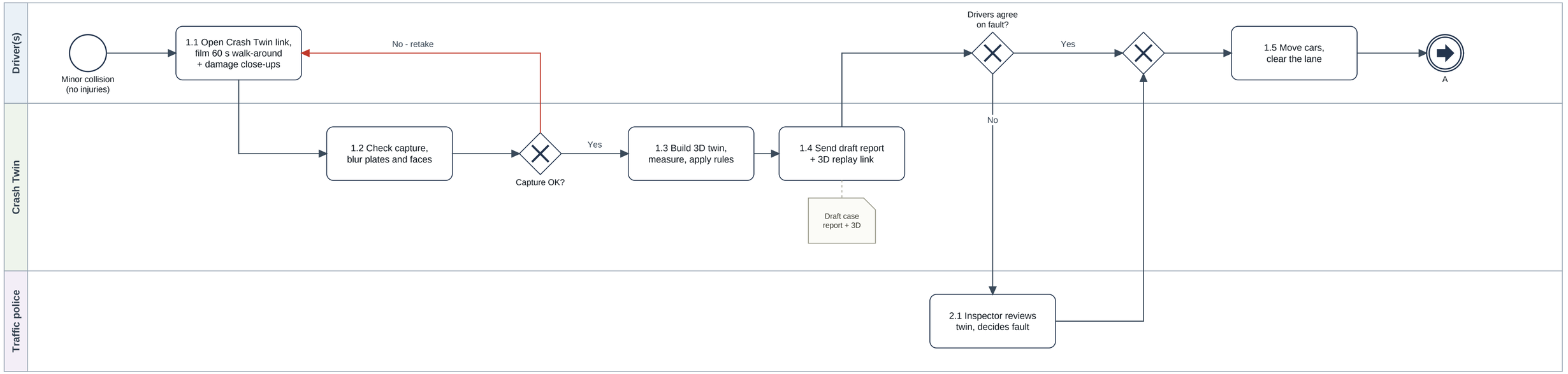
P1 - Business flow: minor collision to settled claim (to-be, with Crash Twin)

Pool 1: at the scene (minutes, lane cleared) | Pool 2: insurance claim (link A continues the case)

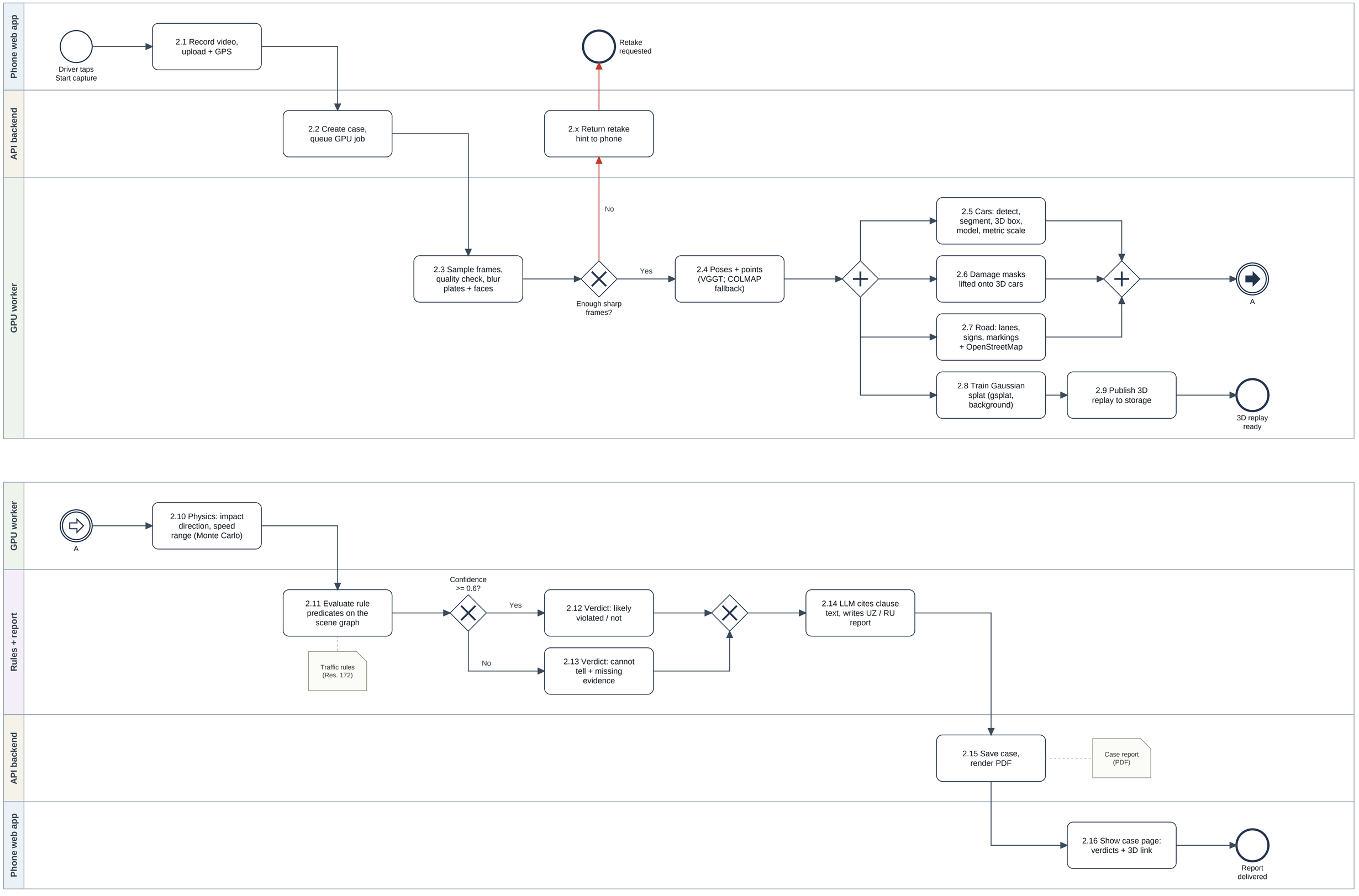
Standards: Road Traffic Rules - CM Resolution No. 172 (12.04.2022) | Fault decision stays with police / court; Crash Twin drafts the evidence



P2 - Technical pipeline: one case, from phone video to rule verdicts

Instant report about 1-3 min; full 3D replay 5-15 min in the background (estimates, to be measured)

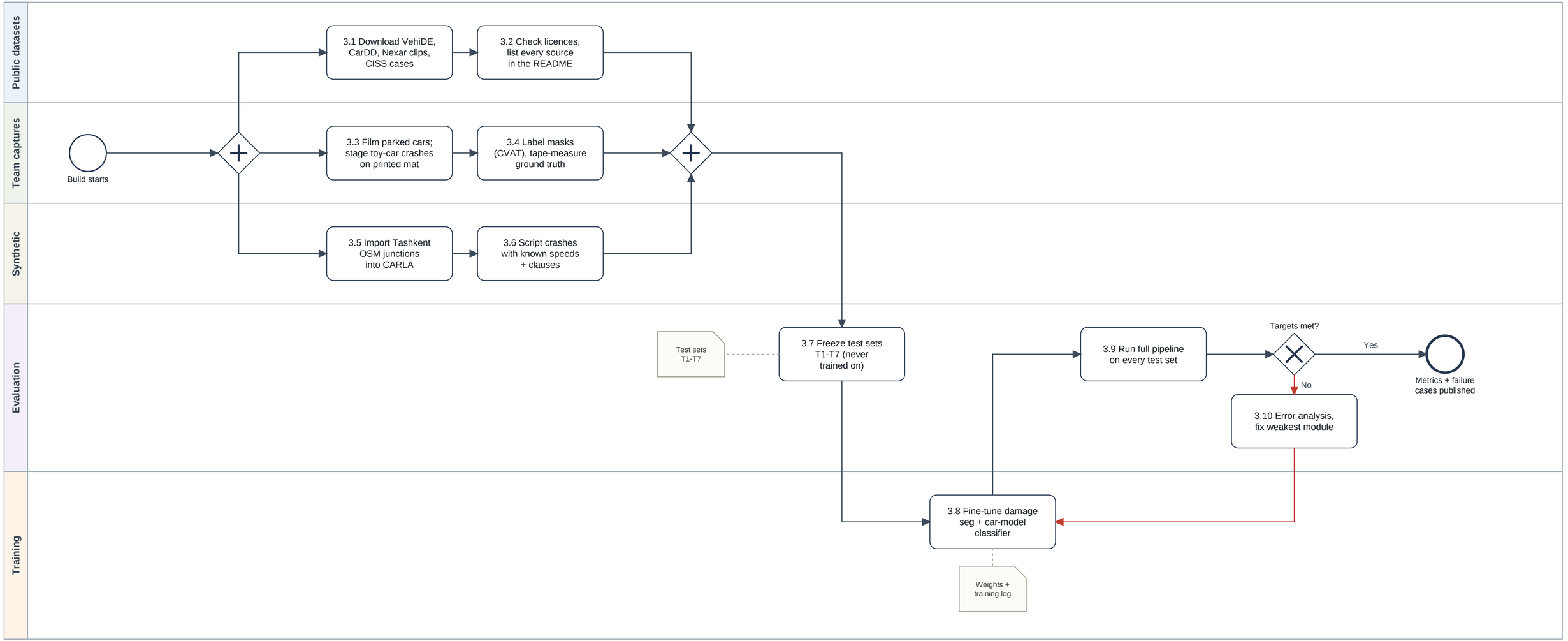
Standards: Models: VGGT-1B-Commercial, COLMAP/GLOMAP, RT-DETR + SAM 2, gsplat | Map: OpenStreetMap (ODbL) | The LLM writes text only, never the verdict



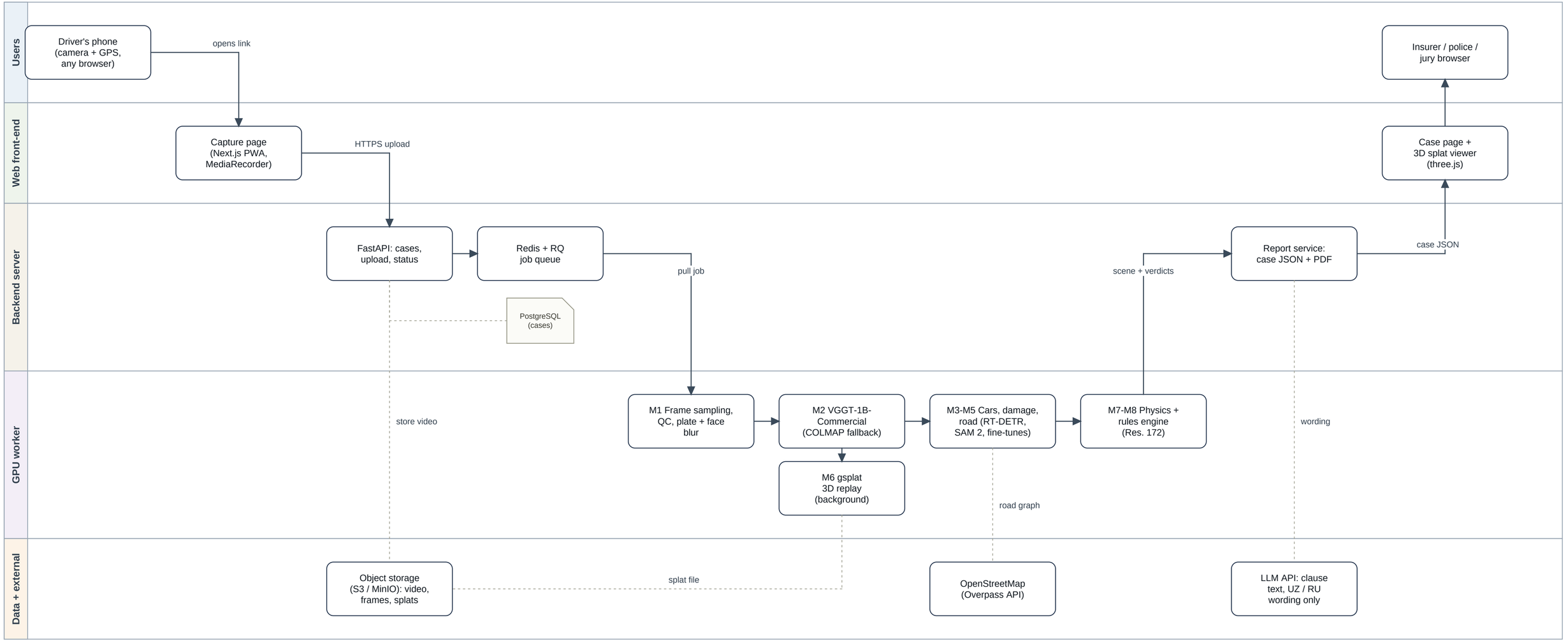
P3 - Data, training and testing flow

Test sets are fixed before training starts, so no reported number is measured on data the model has seen

Standards: Public: VehiDE, CarDD (research), Nexar (Nexar ODL), NHTSA CISS (US public) | Synthetic: CARLA (MIT) on OSM (ODbL) | Own: parked cars + toy crashes, plates/faces blurred



P4 - System architecture and tech stack (demo deployment)

One rented GPU server (RTX 4090 / L4 / A100 class, 24 GB+) runs backend and worker; front-end is static on Vercel; deployed from the public repo

Standards: Python 3, PyTorch, FastAPI, Redis/RQ, PostgreSQL, S3/MinIO | Next.js, three.js splat viewer | VGGT-1B-Commercial, COLMAP/GLOMAP (BSD), RT-DETR (Apache-2.0), SAM 2 (Apache-2.0), gsplat (Apache-2.0)